1 The real partner wins on 180 of 180 never-seen values

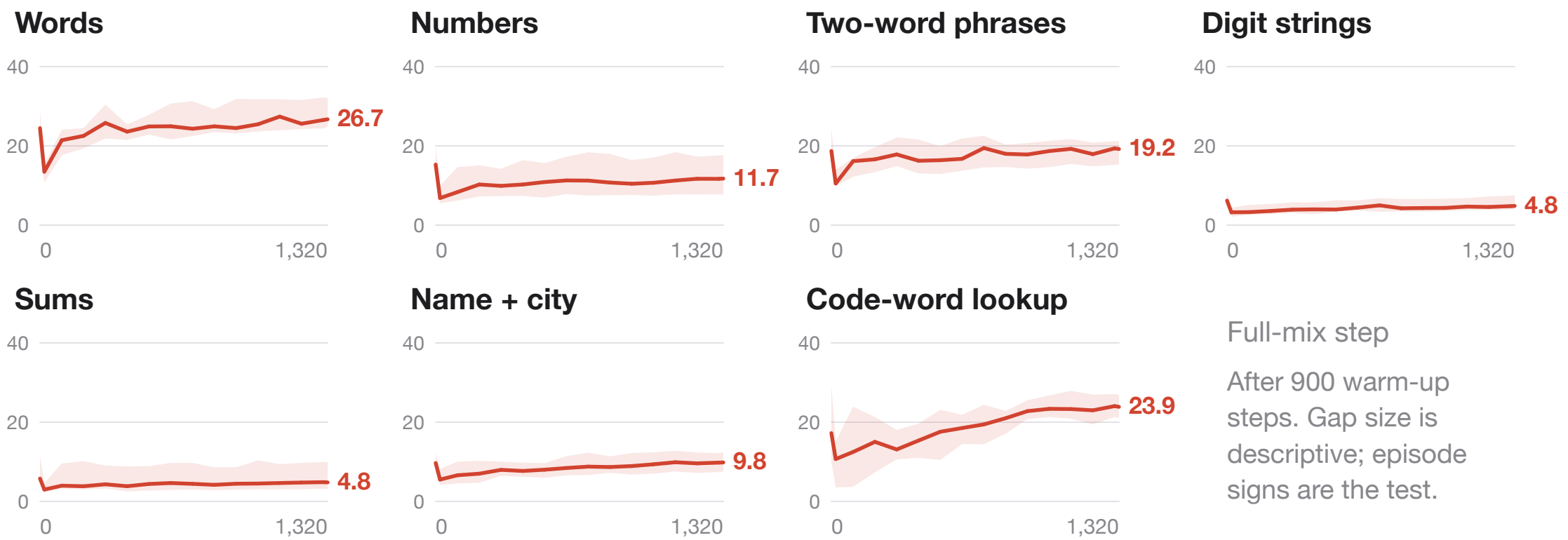
● real partner wins ● shuffled partner wins ○ exact tie (excluded) one dot = one receiver episode

	Into Quill	Into Ember
Words	10/10	10/10
Numbers	10/10	10/10
Two-word phrases	10/10	10/10
Digit strings	10/10	10/10
Sums	20/20	20/20
Name + city	20/20	20/20
Code-word lookup	10/10	10/10

Lower cross-entropy on the payload, compared with a same-role shuffled partner.

2 The advantage survives the full training mix

Median shuffled – real cross-entropy (nats / payload token); shade = middle half of episodes



Held out as values: disjoint word, phrase, name, city and code-word pools.

Numbers / digits: digit sum ≡ 3 mod 7. Sum operands: 50–99 (training: 0–49).

nl20 final · these values may still occur as ordinary words in training text